NPU 开发案例

Revision History

Draft Date	Revision No.	Description
2024/10/22	V1.0	1. 初始版本。

目　　录

前　言 3
1　案例说明 5
2　案例测试 5
3　案例编译 12
　3.1　环境搭建 12
　3.2　案例编译 13
4　关键代码 15
　4.1　mobilenetv1_label_image.cc 15
　4.2　mainwindow.cpp 16
　4.3　main.cpp 17
更多帮助 18

前 言

本文主要介绍基于创龙科技 TLIMX8MP-EVM 评估板的 NPU 开发案例，适用开发环境如下。

Windows 开发环境：Windows 7 64bit、Windows 10 64bit

虚拟机：VMware15.5.5

开发环境：Ubuntu20.04.6 64bit

U-Boot：U-Boot-2022.04

Kernel：Linux-5.15.71-rt51

LinuxSDK：Real-Time Edge Software 2.5

NPU(Neural network Processing Unit)，即神经网络处理器。IMX8MP 内部已集成高能效神经网络处理器 NPU，支持神经网络推理硬件加速，能够流畅运行 AI 算法。主要参数如下：

（1） 支持 INT16/INT32/FP16/FP32；

（2） 支持多种框架，如 TensorFlow Lite/ArmNN/ONNX/DeepViewRT 等；

（3） 2.3TOPS 算力。

mobilenet_v1_1.0_224_quant.tflite 是一个采用 MobileNet V1 架构经过量化处理的 TensorFlow Lite 模型文件，该模型特别适用于在嵌入式设备上执行实时图像分类任务。

MobileNet V1 是一种轻量级的卷积神经网络，专为移动和边缘设备设计，旨在高效地进行图像分类和物体检测。MobileNet 的核心技术是深度可分离卷积，它将标准卷积分解为逐通道卷积和 1x1 卷积，这显著减少了计算和模型大小。尽管模型小巧，但 MobileNet V1 在图像分类上，输出图像中物体的标签及其置信度仍能达到较高的精度。

程序性能参考如下：

（1） 模型输入：模型输入的图像，为程序处理后的 224x224 像素大小，位深为 3 的 RGB 图像；

（2） 平均处理帧率：为模型单张图片循环推理 10 次，每秒可以完成的推理次数（帧率）。

表 1

模型	模型输入	平均处理帧率
Tesorflow lite	224 x 224 x 3 的 RGB 图像	377fps

我司提供的 NPU 开发案例为产品资料“4-软件资料\Demo\platform-demos\mobilenet v1_label_image”，具体说明如下。

表 1

mobilenetv1_label_image	bin	bmp_image	测试图片集
		labels.txt	类别数据集文件
		mobilenet_v1_1.0_224_quant.tflite	tflite模型
		mobilenetv1_label_image	可执行文件
	src		程序源码
	tensorflow-lite_2.9.1-r0.tar.gz		tensorflow-lite 源码压缩包

无特殊说明情况下，本文默认使用 USB TO UART2 作为调试串口，使用系统启动卡（Micro SD 方式）启动系统，通过路由器与 PC 机进行网络连接，请确保 PC 机、Ubuntu 系统可正常访问互联网。

1 案例说明

本案例基于预训练的 TensorFlow Lite 模型实现对图片中目标对象的分类。TensorFlow Lite 模型循环测试 10 次，统计出推理的平均处理耗时和帧率，获取模型输出的前五个标签及置信度打印至串口终端并通过 HDMI 显示屏绘制标签及置信度概率最大的对象结果。

程序处理流程图如下：

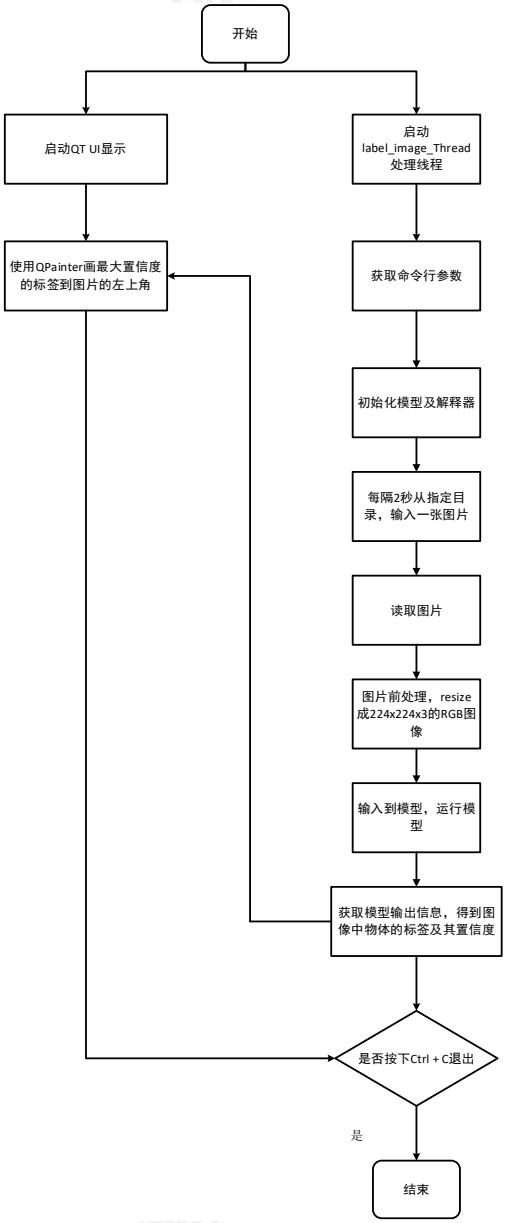

图 1

2 案例测试

请通过网线将评估板千兆网口 ETH0 连接至路由器，将 HDMI 显示器与评估板 HDMI2 OUT 接口(CON22)连接。

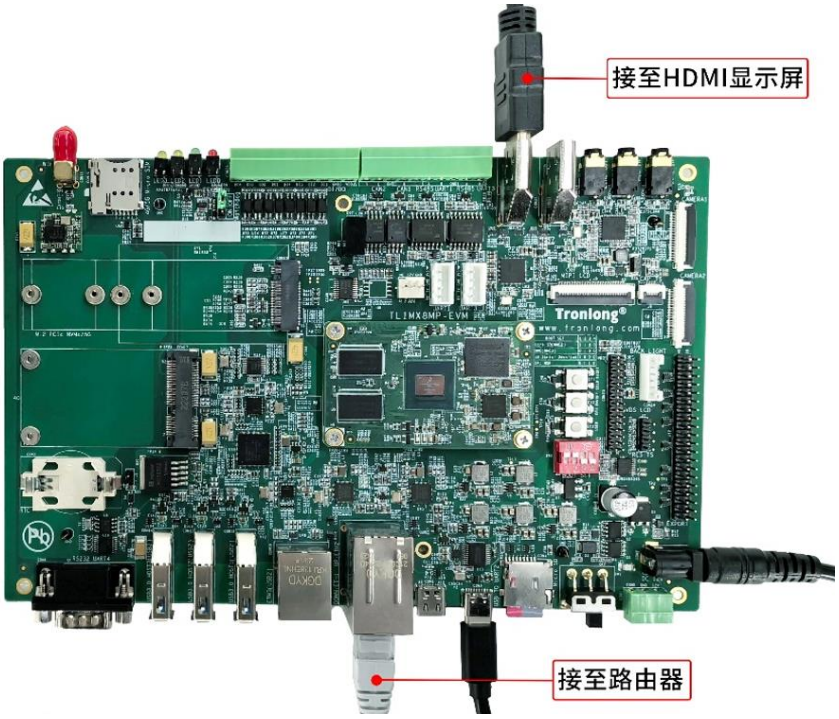

图 2

请将案例 bin 目录下的所有文件拷贝至评估板文件系统任意目录下。

```
root@IMX8MP-Tronlong:~# ls
bmp_image  labels.txt  mobilenet_v1_1.0_224_quant.tflite  mobilenetv1_label_image
root@IMX8MP-Tronlong:~#
```

图 3

在可执行文件所在目录，执行如下命令查看 mobilenetv1_label_image 程序参数说明。

Target# ./mobilenetv1_label_image --help

```
root@IMX8MP-Tronlong:~# ./mobilenetv1_label_image --help
Duplicate flags: num_threads
Duplicate flags: max_delegated_partitions
Duplicate flags: min_nodes_per_partition
Duplicate flags: delegate_serialize_dir
Duplicate flags: delegate_serialize_token
INFO:
usage: label_image <flags>
Flags:
        --num_threads=-1                int32   optional        number of threads used for inference on CPU.
        --max_delegated_partitions=0    int32   optional        Max number of partitions to be delegated.
        --min_nodes_per_partition=0     int32   optional        The minimal number of TFLite graph nodes of a partition that h
as to be reached for it to be delegated.A negative value or 0 means to use the default choice of each delegate.
        --delegate_serialize_dir=       string  optional        Directory to be used by delegates for serializing any model da
ta. This allows the delegate to save data into this directory to reduce init time after the first run. Currently supported by
NNAPI delegate with specific backends on Android. Note that delegate_serialize_token is also required to enable this feature.
        --delegate_serialize_token=     string  optional        Model-specific token acting as a namespace for delegate serial
ization. Unique tokens ensure that the delegate doesn't read inapplicable/invalid data. Note that delegate_serialize_dir is al
so required to enable this feature.
        --num_threads=-1                int32   optional        number of threads used for inference on CPU.
        --max_delegated_partitions=0    int32   optional        Max number of partitions to be delegated.
        --min_nodes_per_partition=0     int32   optional        The minimal number of TFLite graph nodes of a partition that h
as to be reached for it to be delegated.A negative value or 0 means to use the default choice of each delegate.
        --delegate_serialize_dir=       string  optional        Directory to be used by delegates for serializing any model da
ta. This allows the delegate to save data into this directory to reduce init time after the first run. Currently supported by
NNAPI delegate with specific backends on Android. Note that delegate_serialize_token is also required to enable this feature.
        --delegate_serialize_token=     string  optional        Model-specific token acting as a namespace for delegate serial
ization. Unique tokens ensure that the delegate doesn't read inapplicable/invalid data. Note that delegate_serialize_dir is al
so required to enable this feature.
        --external_delegate_path=       string  optional        The library path for the underlying external.
        --external_delegate_options=    string  optional        A list of comma-separated options to be passed to the external
 delegate. Each option is a colon-separated key-value pair, e.g. option_name:option_value.
        --accelerated, -a: [0|1] use Android NNAPI or not
        --allow_fp16, -f: [0|1], allow running fp32 models with fp16 or not
        --count, -c: loop interpreter->Invoke() for certain times
        --gl_backend, -g: [0|1]: use GL GPU Delegate on Android
        --hexagon_delegate, -j: [0|1]: use Hexagon Delegate on Android
        --input_mean, -b: input mean
        --input_std, -s: input standard deviation
        --image, -y: bmp_image directory path
        --labels, -l: labels for the model
        --tflite_model, -m: model_name.tflite
        --profiling, -p: [0|1], profiling or not
        --num_results, -r: number of results to show
        --threads, -t: number of threads
        --verbose, -v: [0|1] print more information
        --warmup_runs, -w: number of warmup runs
        --xnnpack_delegate, -x [0;1]: xnnpack delegate
        --help, -h: Print this help message

qt.qpa.wayland: Creating a fake screen in order for Qt not to crash
xdg_wm_base@16: error 4: xdg_surface geometry (5 x 12) does not match the configured maximized state (0 x 0)
The Wayland connection experienced a fatal error: Protocol error
root@IMX8MP-Tronlong:~#
```

图 4

部分参数解释

（1） --external_delegate_path：动态库路径；

（2） --count：循环测试次数；

（3） --image：测试图片路径；

（4） --labels：类别数据集文件路径；

（5） --tflite_model：tflite 模型路径；

（6） --num_results：识别结果数量；

（7） --verbose：打印模型运行详细信息。

执行如下命令，对图片目标对象进行推理。

Target# ./mobilenetv1_label_image -m mobilenet_v1_1.0_224_quant.tflite -y ./bmp_image -l labels.txt -c 10 --external_delegate_path=/usr/lib/libvx_delegate.so

```
root@IMX8MP-Tronlong:~# ./mobilenetv1_label_image -m mobilenet_v1_1.0_224_quant.tflite -y ./bmp_image -l labels.txt -c 10 --ex
ternal_delegate_path=/usr/lib/libvx_delegate.so
Duplicate flags: num_threads
Duplicate flags: max_delegated_partitions
Duplicate flags: min_nodes_per_partition
Duplicate flags: delegate_serialize_dir
Duplicate flags: delegate_serialize_token
INFO: Loaded model mobilenet_v1_1.0_224_quant.tflite
INFO: resolved reporter
Vx delegate: allowed_cache_mode set to 0.
Vx delegate: device num set to 0.
Vx delegate: allowed_builtin_code set to 0.
Vx delegate: error_during_init set to 0.
Vx delegate: error_during_prepare set to 0.
Vx delegate: error_during_invoke set to 0.
EXTERNAL delegate created.
INFO: Applied EXTERNAL delegate.
W [HandleLayoutInfer:274]Op 162: default layout inference pass.
INFO: invoked
INFO: loop count=10,  average run 2.6519 ms,  377 fps
INFO: 0.537255:  desktop computer
INFO: 0.2:  screen
INFO: 0.0745098:  monitor
INFO: 0.0431373:  notebook
INFO: 0.027451:  desk
INFO: invoked
INFO: loop count=10,  average run 2.6232 ms,  381 fps
INFO: 1:  ping-pong ball
INFO: invoked
INFO: loop count=10,  average run 2.6809 ms,  373 fps
INFO: 1:  airliner
INFO: invoked
INFO: loop count=10,  average run 2.6227 ms,  381 fps
INFO: 0.92549:  palace
INFO: 0.0470588:  monastery
INFO: 0.0117647:  bell cote
INFO: 0.00392157:  umbrella
INFO: 0.00392157:  church
INFO: invoked
INFO: loop count=10,  average run 2.5975 ms,  384 fps
INFO: 0.623529:  printer
INFO: 0.380392:  photocopier
INFO: invoked
INFO: loop count=10,  average run 2.6524 ms,  377 fps
INFO: 0.996078:  Tibetan mastiff
INFO: 0.00392157:  chow
INFO: 0.00392157:  Newfoundland
INFO: invoked
INFO: loop count=10,  average run 2.6695 ms,  374 fps
INFO: 1:  sundial
```

图 5

```
INFO: invoked
INFO: loop count=10,  average run 2.559 ms,  390 fps
INFO: 0.8:  computer keyboard
INFO: 0.129412:  space bar
INFO: 0.0196078:  mouse
INFO: 0.0196078:  abacus
INFO: 0.0117647:  typewriter keyboard
INFO: invoked
INFO: loop count=10,  average run 2.6481 ms,  377 fps
INFO: 0.537255:  desktop computer
INFO: 0.2:  screen
INFO: 0.0745098:  monitor
INFO: 0.0431373:  notebook
INFO: 0.027451:  desk
INFO: invoked
INFO: loop count=10,  average run 2.5854 ms,  386 fps
INFO: 1:  ping-pong ball
```

图 6

参数解释

（1） -m：tflite 模型的路径；

（2） -y：测试图片的路径；

（3） -l：类别数据集文件的路径；

（4） -c：循环测试的次数；

（5） --external_delegate_path：通过 NNAPI Delegate 在 NPU 上加速模型运行。

程序滚动识别 bmp_image 文件夹内的图片，以第 1 张图片识别结果为例，输出结果信息如下所示。

```
INFO: invoked
INFO: loop count=10,  average run 2.6519 ms,  377 fps
INFO: 0.537255:  desktop computer
INFO: 0.2:  screen
INFO: 0.0745098:  monitor
INFO: 0.0431373:  notebook
INFO: 0.027451:  desk
```

图 7

从输出信息可知，本张图片识别为桌面电脑、屏幕、监控器、笔记本电脑、桌子的概率分别为 53.7255%、20%、7.45098%、4.31373%、2.7451%，程序循环运行 10 次模型平均耗时为 2.6519ms，帧率为 377fps。

案例程序对测试图片的目标对象进行识别后，通过 HDMI 显示屏绘制标签及置信度概率最大的对象结果如下所示。

图 8

评估板默认支持 HDMI1 OUT 和 HDMI2 OUT 接口显示，其中 HDMI1 OUT 为副显示，HDMI2 OUT 为主显示。如需使用 HDMI1 OUT 接口进行测试，请按照如下步骤修改内容，再运行案例可执行程序。否则请忽略此内容。

请执行如下命令，修改"/etc/xdg/weston/weston.ini"文件，将 HDMI2 OUT 内部输出关闭。

Target# vi /etc/xdg/weston/weston.ini

```
root@IMX8MP-Tronlong:~# vi /etc/xdg/weston/weston.ini
root@IMX8MP-Tronlong:~#
```

图 9

添加内容如下：

[output]

name=DSI-1

mode=off

```
[core]
#gbm-format=argb8888
idle-time=0
use-g2d=1
#xwayland=true
repaint-window=16
#enable-overlay-view=1
modules=screen-share.so

#[shell]
#size=1920x1080

[libinput]
touchscreen_calibrator=true

[output]
name=DSI-1
mode=off

#[output]
#name=HDMI-A-1
#mode=1920x1080@60
#transform=rotate-90

#[output]
#name=HDMI-A-2
#mode=off
#       WIDTHXHEIGHT    Resolution size width and height in pixels
#       off             Disables the output
#       preferred       Uses the preferred mode
#       current         Uses the current crt controller mode
#transform=rotate-90

[screen-share]
command=/usr/bin/weston --backend=rdp-backend.so --shell=fullscreen-shell.so --no-clients-resize --rdp-tls-cert=/etc/freerdp/k
#start-on-startup=true
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
I /etc/xdg/weston/weston.ini [Modified] 18/38 47%
```

图 10

修改完后保存文件，执行如下命令重启 Weston 服务。

Target# systemctl restart weston

```
root@IMX8MP-Tronlong:~# systemctl restart weston
[ 2965.133325] audit: type=1006 audit(1651184699.392:4): pid=884 uid=0 old-auid=4294967295 auid=0 tty=tty7 old-ses=4294967295
ses=3 res=1
[ 2965.133344] audit: type=1300 audit(1651184699.392:4): arch=c00000b7 syscall=64 success=yes exit=1 a0=8 a1=ffffe64fcd40 a2=1
 a3=0 items=0 ppid=1 pid=884 auid=0 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=tty7 ses=3 comm="(weston)" exe
="/lib/systemd/systemd" key=(null)
[ 2965.133352] audit: type=1327 audit(1651184699.392:4): proctitle="(weston)"
root@IMX8MP-Tronlong:~#
```

图 11

重启 Weston 服务后执行如下命令，对图片目标对象进行推理。

Target# ./mobilenetv1_label_image -m mobilenet_v1_1.0_224_quant.tflite -y ./bmp_image -l labels.txt -c 10 --external_delegate_path=/usr/lib/libvx_delegate.so

```
root@IMX8MP-Tronlong:~# ./mobilenetv1_label_image -m mobilenet_v1_1.0_224_quant.tflite -y ./bmp_image -l labels.txt -c 10 --ex
ternal_delegate_path=/usr/lib/libvx_delegate.so
Duplicate flags: num_threads
Duplicate flags: max_delegated_partitions
Duplicate flags: min_nodes_per_partition
Duplicate flags: delegate_serialize_dir
Duplicate flags: delegate_serialize_token
INFO: Loaded model mobilenet_v1_1.0_224_quant.tflite
INFO: resolved reporter
Vx delegate: allowed_cache_mode set to 0.
Vx delegate: device num set to 0.
Vx delegate: allowed_builtin_code set to 0.
Vx delegate: error_during_init set to 0.
Vx delegate: error_during_prepare set to 0.
Vx delegate: error_during_invoke set to 0.
EXTERNAL delegate created.
INFO: Applied EXTERNAL delegate.
W [HandleLayoutInfer:274]Op 162: default layout inference pass.
INFO: invoked
INFO: loop count=10,  average run 2.6519 ms,  377 fps
INFO: 0.537255:  desktop computer
INFO: 0.2:  screen
INFO: 0.0745098:  monitor
INFO: 0.0431373:  notebook
INFO: 0.027451:  desk
INFO: invoked
INFO: loop count=10,  average run 2.6232 ms,  381 fps
INFO: 1:  ping-pong ball
INFO: invoked
INFO: loop count=10,  average run 2.6809 ms,  373 fps
INFO: 1:  airliner
INFO: invoked
INFO: loop count=10,  average run 2.6227 ms,  381 fps
INFO: 0.92549:  palace
INFO: 0.0470588:  monastery
INFO: 0.0117647:  bell cote
INFO: 0.00392157:  umbrella
INFO: 0.00392157:  church
INFO: invoked
INFO: loop count=10,  average run 2.5975 ms,  384 fps
INFO: 0.623529:  printer
INFO: 0.380392:  photocopier
INFO: invoked
INFO: loop count=10,  average run 2.6524 ms,  377 fps
INFO: 0.996078:  Tibetan mastiff
INFO: 0.00392157:  chow
INFO: 0.00392157:  Newfoundland
INFO: invoked
INFO: loop count=10,  average run 2.6695 ms,  374 fps
INFO: 1:  sundial
```

图 12

```
INFO: invoked
INFO: loop count=10,  average run 2.559 ms,  390 fps
INFO: 0.8:  computer keyboard
INFO: 0.129412:  space bar
INFO: 0.0196078:  mouse
INFO: 0.0196078:  abacus
INFO: 0.0117647:  typewriter keyboard
INFO: invoked
INFO: loop count=10,  average run 2.6481 ms,  377 fps
INFO: 0.537255:  desktop computer
INFO: 0.2:  screen
INFO: 0.0745098:  monitor
INFO: 0.0431373:  notebook
INFO: 0.027451:  desk
INFO: invoked
INFO: loop count=10,  average run 2.5854 ms,  386 fps
INFO: 1:  ping-pong ball
```

图 13

3 案例编译

3.1 环境搭建

将案例 tensorflow-lite 源码包拷贝至 Ubuntu 工作目录下，执行如下命令进行解压。

Host# tar -xzf tensorflow-imx.tar.gz

```
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image$ ls
bin  src  tensorflow-lite_2.9.1-r0.tar.gz
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image$ tar -xzf tensorflow-lite_2.9.1-r0.tar.gz
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image$
```

图 14

解压完成后进入工作目录下的 2.9.1-r0 目录，执行如下命令使能交叉编译工具链。

Host# source /home/tronlong/SDK/environment-setup-armv8a-poky-linux

备注：SDK 安装路径请以实际路径为准。

```
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image$ cd 2.9.1-r0/
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/2.9.1-r0$ ls
build                                                package
deploy-debs                                          packages-split
deploy-source-date-epoch                             pkgdata
git                                                  pkgdata-pdata-input
image                                                pkgdata-sysroot
license-destdir                                      pseudo
mobilenet_v1_1.0_224_quant.ckpt.data-00000-of-00001  recipe-sysroot
mobilenet_v1_1.0_224_quant.ckpt.index                recipe-sysroot-native
mobilenet_v1_1.0_224_quant.ckpt.meta                 source-date-epoch
mobilenet_v1_1.0_224_quant_eval.pbtxt                sysroot-destdir
mobilenet_v1_1.0_224_quant_frozen.pb                 temp
mobilenet_v1_1.0_224_quant_info.txt                  tensorflow-lite.pc.in
mobilenet_v1_1.0_224_quant.tflite                    toolchain.cmake
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/2.9.1-r0$ source /home/tronlong
/SDK/environment-setup-armv8a-poky-linux
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/2.9.1-r0$
```

图 15

3.2 案例编译

在案例 src 源码目录下，执行如下命令修改 tensorflow-lite 源码包解压安装路径。

备注：请根据解压的实际安装路径进行修改。

Host# vim mobilenetv1_label_image.pro

```
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/src$ ls
bitmap_helpers.cc      get_top_n.h        main.cpp         mobilenetv1_label_image.cc
bitmap_helpers.h       get_top_n_impl.h   mainwindow.cpp   mobilenetv1_label_image.h
bitmap_helpers_impl.h  log.h              mainwindow.h     mobilenetv1_label_image.pro
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/src$ vim mobilenetv1_label_imag
e.pro
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/src$
```

图 16

修改内容如下：

[TENSORFLOW_SOURCE_DIR = /home/tronlong/Demo/platform-demos/mobilenetv1_label_image/2.9.1-r0

```
 8 TARGET = mobilenetv1_label_image
 9 TEMPLATE = app
10
11 DEFINES += QT_DEPRECATED_WARNINGS
12
13 TENSORFLOW_SOURCE_DIR = /home/tronlong/Demo/platform-demos/mobilenetv1_label_image/2.9.1-r0
14 TENSORFLOW_SOURCE_DIR_BUILD = $${TENSORFLOW_SOURCE_DIR}/build/abseil-cpp
15 TENSORFLOW_SOURCE_DIR_GIT = $${TENSORFLOW_SOURCE_DIR}/git
16 TF_SOURCE_DIR = $${TENSORFLOW_SOURCE_DIR_GIT}/tensorflow
17 TFLITE_SOURCE_DIR = $${TENSORFLOW_SOURCE_DIR_GIT}/tensorflow/lite
18
19 SOURCES += \
20     bitmap_helpers.cc \
21     mobilenetv1_label_image.cc \
```

图 17

修改保存退出后，执行如下命令生成 Makefile 文件。

Host# qmake

```
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/src$ qmake
Info: creating stash file /home/tronlong/Demo/platform-demos/mobilenetv1_label_image/src/.qmake.stash
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/src$ ls
bitmap_helpers.cc      get_top_n_impl.h   mainwindow.h                mobilenetv1_label_image.pro
bitmap_helpers.h       log.h              Makefile
bitmap_helpers_impl.h  main.cpp           mobilenetv1_label_image.cc
get_top_n.h            mainwindow.cpp     mobilenetv1_label_image.h
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/src$
```

图 18

执行如下命令进行案例编译，编译完成将会在目录下生成可执行文件。

Host# make

备注：编译过程生成的警告可忽略，不影响案例功能。

```
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/src$ make
aarch64-poky-linux-g++  -march=armv8-a+crc+crypto -fstack-protector-strong  -O2 -D_FORTIFY_SOURCE=2 -Wformat -W
format-security -Werror=format-security --sysroot=/home/tronlong/SDK/sysroots/armv8a-poky-linux -c -pipe  -O2 -
pipe -g -feliminate-unused-debug-types  -O2 -Wall -Wextra -D_REENTRANT -fPIC -DQT_DEPRECATED_WARNINGS -DQT_NO_D
EBUG -DQT_PRINTSUPPORT_LIB -DQT_WIDGETS_LIB -DQT_SVG_LIB -DQT_GUI_LIB -DQT_CONCURRENT_LIB -DQT_CORE_LIB -I. -I.
./2.9.1-r0/git -I../2.9.1-r0/build/abseil-cpp -I. -I../../../../SDK/sysroots/armv8a-poky-linux/usr/include/QtPr
intSupport -I../../../../SDK/sysroots/armv8a-poky-linux/usr/include/QtWidgets -I../../../../SDK/sysroots/armv8a
-poky-linux/usr/include/QtSvg -I../../../../SDK/sysroots/armv8a-poky-linux/usr/include/QtGui -I../../../../SDK/
sysroots/armv8a-poky-linux/usr/include/QtConcurrent -I../../../../SDK/sysroots/armv8a-poky-linux/usr/include/Qt
Core -I. -I../../../../SDK/sysroots/armv8a-poky-linux/usr/lib/mkspecs/linux-oe-g++ -o bitmap_helpers.o bitmap_h
elpers.cc
In file included from ../2.9.1-r0/git/tensorflow/lite/interpreter_builder.h:30,
                 from ../2.9.1-r0/git/tensorflow/lite/model.h:21,
                 from mobilenetv1_label_image.h:19,
                 from bitmap_helpers_impl.h:19,
                 from bitmap_helpers.h:19,
                 from bitmap_helpers.cc:16:
../2.9.1-r0/git/tensorflow/lite/core/api/op_resolver.h: In member function 'virtual tflite::OpResolver::TfLiteD
elegatePtrVector tflite::OpResolver::GetDelegates(int) const':
../2.9.1-r0/git/tensorflow/lite/core/api/op_resolver.h:46:52: warning: unused parameter 'num_threads' [-Wunused
-parameter]
   46 |   virtual TfLiteDelegatePtrVector GetDelegates(int num_threads) const {
      |                                                   ~~~~~^~~~~~~~~~~
In file included from ../2.9.1-r0/git/tensorflow/lite/core/subgraph.h:32,
                 from ../2.9.1-r0/git/tensorflow/lite/interpreter_builder.h:31,
                 from ../2.9.1-r0/git/tensorflow/lite/model.h:21,
                 from mobilenetv1_label_image.h:19,
                 from bitmap_helpers_impl.h:19,
                 from bitmap_helpers.h:19,
                 from bitmap_helpers.cc:16:
../2.9.1-r0/git/tensorflow/lite/core/api/profiler.h: In member function 'virtual void tflite::Profiler::EndEven
t(uint32_t, int64_t, int64_t)':
```

图 19

```
../2.9.1-r0/git/tensorflow/lite/core/subgraph.h: In member function 'virtual void tflite::Subgraph::SubgraphAwa
reProfiler::AddEvent(const char*, tflite::Profiler::EventType, uint64_t, uint64_t, int64_t, int64_t)':
../2.9.1-r0/git/tensorflow/lite/core/subgraph.h:430:27: warning: unused parameter 'event_metadata2' [-Wunused-p
arameter]
  430 |                    int64_t event_metadata2) override {
      |                    ~~~~~~~~^~~~~~~~~~~~~~~
aarch64-poky-linux-g++  -march=armv8-a+crc+crypto -fstack-protector-strong  -O2 -D_FORTIFY_SOURCE=2 -Wformat -W
format-security -Werror=format-security --sysroot=/home/tronlong/SDK/sysroots/armv8a-poky-linux -Wl,-O1 -Wl,--h
ash-style=gnu -Wl,--as-needed  -Wl,-z,relro,-z,now -Wl,-O1 -Wl,-rpath-link,/home/tronlong/SDK/sysroots/armv8a-p
oky-linux/usr/lib -o mobilenetv1_label_image  bitmap_helpers.o mobilenetv1_label_image.o stats_calculator.o mem
ory_info.o profile_summarizer.o profile_summary_formatter.o time.o command_line_flags.o default_execution_provi
der.o delegate_provider.o utils.o tool_params.o external_delegate_provider.o mainwindow.o main.o moc_mainwindow
.o   -ltensorflow-lite -lrt /home/tronlong/SDK/sysroots/armv8a-poky-linux/usr/lib/libQt6PrintSupport.so /home/t
ronlong/SDK/sysroots/armv8a-poky-linux/usr/lib/libQt6Widgets.so /home/tronlong/SDK/sysroots/armv8a-poky-linux/u
sr/lib/libQt6Svg.so /home/tronlong/SDK/sysroots/armv8a-poky-linux/usr/lib/libGLESv2.so /home/tronlong/SDK/sysro
ots/armv8a-poky-linux/usr/lib/libEGL.so /home/tronlong/SDK/sysroots/armv8a-poky-linux/usr/lib/libQt6Gui.so -lEG
L /home/tronlong/SDK/sysroots/armv8a-poky-linux/usr/lib/libQt6Concurrent.so /home/tronlong/SDK/sysroots/armv8a-
poky-linux/usr/lib/libQt6Core.so -lpthread -lGLESv2
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/src$ ls
bitmap_helpers.cc             get_top_n.h        Makefile                    moc_mainwindow.o
bitmap_helpers.h              get_top_n_impl.h   memory_info.o               moc_predefs.h
bitmap_helpers_impl.h         log.h              mobilenetv1_label_image     profile_summarizer.o
bitmap_helpers.o              main.cpp           mobilenetv1_label_image.cc  profile_summary_formatter.o
command_line_flags.o          main.o             mobilenetv1_label_image.h   stats_calculator.o
default_execution_provider.o  mainwindow.cpp     mobilenetv1_label_image.o   time.o
delegate_provider.o           mainwindow.h       mobilenetv1_label_image.pro tool_params.o
external_delegate_provider.o  mainwindow.o       moc_mainwindow.cpp          utils.o
tronlong@tronlong-virtual-machine:~/Demo/platform-demos/mobilenetv1_label_image/src$
```

图 20

4 关键代码

4.1 mobilenetv1_label_image.cc

（1） 初始化代理。

```
int Main(int argc, char* argv[]) {

  DelegateProviders delegate_providers;

  bool parse_result = delegate_providers.InitFromCmdlineArgs(
      &argc, const_cast<const char**>(argv));
  if (!parse_result) {
    display_usage(delegate_providers);
    return EXIT_FAILURE;
  }
```

图 21

（2） 获取命令行参数、初始化参数。

```
  Settings s;
  get_util(s, argc, argv);
```

图 22

（3） 根据传入的设置's'，进行相应的委托设置。

```
  delegate_providers.MergeSettingsIntoParams(s);
```

图 23

（4） 初始化模型和解释器。

```
  // Init the Inference
  delegate_providers.InitInference(delegate_providers, &s);
```

图 24

（5） 运行模型。

```
// Run the Inference
while(! g_quit){
for (const auto& result : files) {
        s.input_bmp_name = s.directory_path + "/" + result;
        delegate_providers.RunInference(&s);
        sleep(2);
    }
}
```

图 25

（6） 释放资源。

```
// Release the Interpreter
delegate_providers.ReleaseInterpreter();
start = false;
std::cout << " delegate_providers.ReleaseInterpreter() " << std::endl;
```

图 26

4.2 mainwindow.cpp

（1） 初始化 Mainwindow。

```
MainWindow::MainWindow(QWidget *parent)
    : QMainWindow(parent) {

    timer = new QTimer(this);
    connect(timer, &QTimer::timeout, this, &MainWindow::checkAndDisplayImage);

    timer->start(100);

    imageLabel = new QLabel(this);
    imageLabel->setAlignment(Qt::AlignCenter);
    setCentralWidget(imageLabel);

}
```

图 27

（2） 显示图片并绘制标签。

```
void MainWindow::displayImage(const QString &imagePath) {

    QPixmap originalPixmap(imagePath);

    if (originalPixmap.isNull()) {
        qWarning("Failed to load image from path: %s", imagePath.toUtf8().constData());
        return;
    }

    // Create a QPixmap of the same size as the original image
    QPixmap pixmap(originalPixmap.size());
    pixmap.fill(Qt::transparent); // Fill with a transparent background

    // Use QPainter to draw the original image and text on the new pixmap
    QPainter painter(&pixmap);
    painter.drawPixmap(0, 0, originalPixmap); // Draw the original image first

    // Sets the font and text position
    QFont font("Arial", 22);
    painter.setFont(font);
    painter.setPen(Qt::red);
    QString fixedValue = QString::number(confidence_, 'f', 1);
    QString text = QString::fromStdString(labels_name);// The displayed text
    QString combined = text + " : " + fixedValue + "%";

    painter.drawText(40, 40 , combined); // Drawing text

    // Displays an image with text
    imageLabel->setPixmap(pixmap);
    imageLabel->resize(pixmap.size());

}
```

图 28

4.3 main.cpp

（1） 主程序入口，创建 label_image_Thread 线程并启动，同时启动 Mainwindow UI 界面显示。

```
int main(int argc, char *argv[]) {

    // Create and start label_image thread
    Label_image label_image_Thread(argc, argv);
    label_image_Thread.start();

    QApplication app(argc, argv);
    MainWindow w;
    w.setWindowState(Qt::WindowMaximized);
    w.setWindowFlag(Qt::FramelessWindowHint);
    w.show();
    app.exec();

    g_quit = true;
    label_image_Thread.quit();
    label_image_Thread.wait();

    return 0;
}
```

图 29

更多帮助

销售邮箱：sales@tronlong.com

技术邮箱：support@tronlong.com

创龙总机：020-8998-6280

技术热线：020-3893-9734

创龙官网：www.tronlong.com

技术论坛：www.51ele.net

官方商城：tronlong.tmall.com